

Scaling Power-Efficient AI Factories with NVIDIA Spectrum-X Ethernet Photonics

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Communications English Scaling Power-Efficient AI Factories with NVIDIA Spectrum-X Ethernet Photonics Jan 06,
2026 By Ashkan Seyedi Like Discuss (0) L T F R E AI-Generated Summary Like Dislike Spectrum-X Ethernet Photonics,
integrated into the NVIDIA Rubin platform, delivers co-packaged optics and silicon photonic engines with 5x power reduction
per 1.6 Tb/s port and 5x longer link flap-free uptime compared to off-the-shelf Ethernet, supporting multi-trillion-parameter AI
factories. The switch system features a fully integrated 512 lane 200G-capable architecture, a detachable fiber connector for
automated large-scale assembly, and a solder-reflow compatible optical engine enabling 100% yield through pre-attachment
component screening and pick-and-place automation. An integrated fiber shuffle mechanism within quad-ASIC SN6800
switches provides 409.6 Tb/s total bandwidth, flat scaling for GPU clusters, and 10x greater resiliency, establishing robust,
energy-efficient networking for expansive, mission-critical AI workloads. AI-generated content may summarize information
incompletely. Verify important information. Learn more NVIDIA is bringing the world's first optimized Ethernet networking with
co-packaged optics to AI factories, enabling scale-out and scale-across on the NVIDIA Rubin platform with NVIDIA
Spectrum-X Ethernet Photonics, the flagship switch for multi-trillion-parameter AI infrastructure. This blog post explores key
optimizations and innovations in the protocol and hardware of Spectrum-X Ethernet Photonics that enable power-efficient,
reliable, and resilient co-packaged optical networks for giga-scale AI factories. How Ethernet for AI enables scalable training
and inference on the NVIDIA Rubin Platform Ultra-low-jitter Ethernet networking plays a vital role in scaling AI factories, as it
ensures consistent and reliable data transmission across the entire infrastructure. By minimizing jitter, AI systems can
achieve efficient token throughput regardless of batch size, which is crucial for handling diverse and demanding workloads.
This ability supports seamless multi-tenancy within a single AI factory, for multiple users and applications to operate
concurrently without performance degradation. It also improves the dispatch efficiency of models based on the Mixture of
Experts (MoE) architecture, enabling faster expert selection and improved overall model performance, as shown in Figure 1.
As a result, AI factories can operate at greater speed, reliability, and scalability. Figure 1. NVIDIA Spectrum-X Ethernet
provides low-jitter communication and higher NVIDIA Collective Communication Library (NCCL) performance over
off-the-shelf Ethernet Key innovations in Spectrum-X Ethernet Photonics for AI factory optical interconnects The Spectrum-X
Ethernet Photonics switch delivers performance improvements for AI factories through its co-packaged silicon photonic
engines. New packaging and low-loss electro-optical channels offer 5x power reduction per 1.6 Tb/s port compared to
pluggable interconnects. The co-packaged optical links sustain 5x longer link flap-free AI uptime compared to off-the-shelf
Ethernet solutions, ensuring AI workloads run without interruption. 10x greater network resiliency provides unmatched
robustness for mission-critical applications. With these innovations, organizations can scale their AI infrastructure and
increase performance per watt, supporting larger workloads while maintaining optimal energy efficiency, reliability, and
network stability. Figure 2. Spectrum-X Ethernet Photonics MCM package Spectrum-X Ethernet Photonics is the world's first
fully integrated 512 lane 200G-capable co-packaged switch system. The introduction of the detachable fiber connector for
surface-normal input/output (I/O) is an advancement in the assembly and scalability of high-performance Ethernet switches
for AI factories. By enabling a fully automated process where optical fibers are attached at the final stage using precision
machinery, manufacturers can maximize production yield and throughput, streamlining large-scale deployment. The
surface-normal optical I/O architecture enables optical ports to scale without increasing the physical size of the switch
package. This is especially advantageous for high radix switches, which require numerous connections within a compact
footprint to support expansive AI workloads. The solder-reflow compatible optical engine is also a breakthrough that
integrates seamlessly with modern test and assembly tools. This compatibility enables full screening of optical components
before attachment to the switch silicon, ensuring that only known-good engines are used, achieving a guaranteed 100%
yield. The process benefits from pick-and-place automation and comprehensive pre-assembly testing, which together
provide an efficient manufacturing pathway for these advanced switch systems. The integrated shuffle mechanism within the
quad-ASIC switch architectures is another key innovation, enabling flat and efficient scaling of GPUs within a single cluster.
This topology eliminates the latency typically introduced by additional switching layers, maintaining optimal performance as
clusters grow. The SN6800 switch delivers 409.6 Tb/s of total bandwidth across 512 ports of 800 Gb/s, or 2,048 ports of 200
Gb/s, using its integrated fiber shuffle and co-packaged silicon photonics to establish a space- and power-efficient Ethernet

solution. These combined innovations equip AI factories with robust, scalable network infrastructure capable of supporting next-generation artificial intelligence applications. Figure 3. Spectrum-X Ethernet Photonics-based SN6800 and SN6810 Ethernet switches What's next for AI factory networking innovation This holistic codesign approach—with chips, systems, software, and AI models—enables the development of scalable, high-performance AI factories. Spectrum-X Ethernet Photonics switches deliver ultra-low jitter networking for AI factories to grow in speed, reliability, and scalability, and establish robust infrastructure for next-generation applications. For more information, see the NVIDIA Silicon Photonics page. Discuss (0) Like Tags Agentic AI / Generative AI | Data Center / Cloud | Networking / Communications | Hardware / Semiconductor | Spectrum-X Ethernet | Intermediate Technical | Deep dive | CES26 | featured | Network Architecture About the Authors